

## Configuring Network Interfaces (ifconfig, ip, nmcli)? Understanding DNS, DHCP, and Static IP Configuration? Firewall Management (iptables, firewalld, ufw)

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**Q.1 Use the ifconfig or ip command to display the current IP address and network interfaces on your Linux system, then take a screenshot of the output.**

**Ans-** Step 1- Show ip address: ip addr show

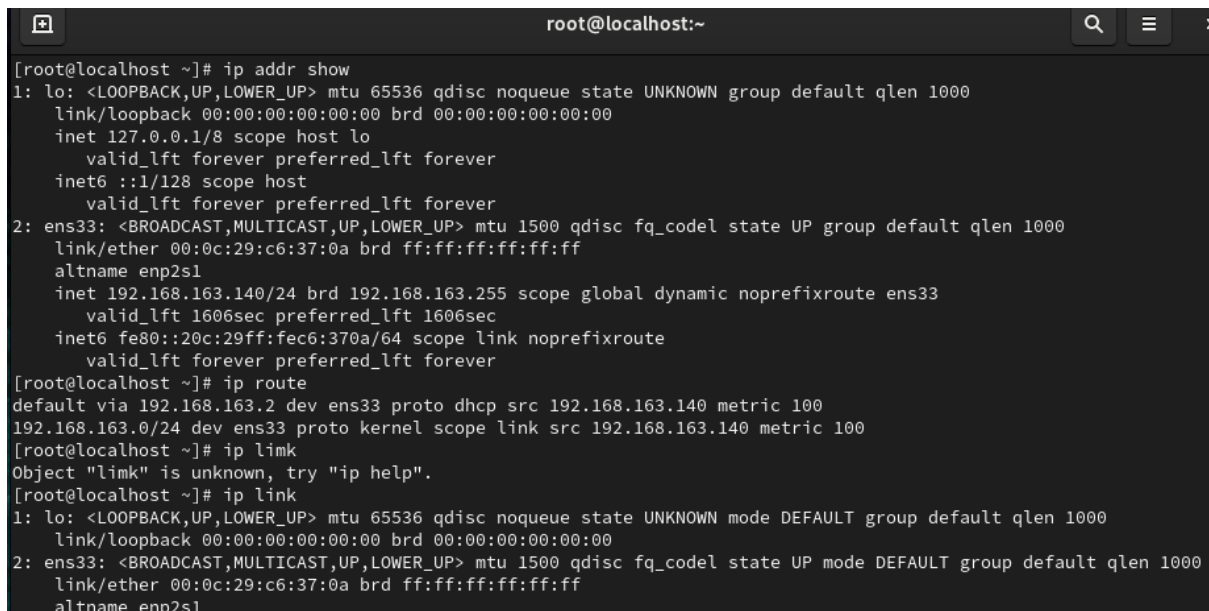
Step 2-Display ip route: ip route

Step 3-Display Interface status: ip link

Step 4-ifconfig

Step 5-sudo dnf install net-tools

Step 6-ifconfig



```
root@localhost:~  
[root@localhost ~]# ip addr show  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:c6:37:0a brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    inet 192.168.163.140/24 brd 192.168.163.255 scope global dynamic noprefixroute ens33  
        valid_lft 1606sec preferred_lft 1606sec  
    inet6 fe80::20c:29ff:fec6:370a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
[root@localhost ~]# ip route  
default via 192.168.163.2 dev ens33 proto dhcp src 192.168.163.140 metric 100  
192.168.163.0/24 dev ens33 proto kernel scope link src 192.168.163.140 metric 100  
[root@localhost ~]# ip link  
Object "link" is unknown, try "ip help".  
[root@localhost ~]# ip link  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000  
    link/ether 00:0c:29:c6:37:0a brd ff:ff:ff:ff:ff:ff  
    altname enp2s1
```

```
[root@localhost ~]# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.163.140 netmask 255.255.255.0 broadcast 192.168.163.255
    inet6 fe80::20c:29ff:fec6:370a prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:c6:37:0a txqueuelen 1000 (Ethernet)
    RX packets 95 bytes 9429 (9.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 90 bytes 9087 (8.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 18 bytes 2112 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 18 bytes 2112 (2.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@localhost ~]# sudo dnf install net-tools
Last metadata expiration check: 0:53:35 ago on Mon 29 Jun 2026 12:39:00 AM EDT.
Package net-tools-2.0-0.64.20160912git.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!

[root@localhost ~]# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.163.140 netmask 255.255.255.0 broadcast 192.168.163.255
    inet6 fe80::20c:29ff:fec6:370a prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:c6:37:0a txqueuelen 1000 (Ethernet)
    RX packets 100 bytes 9849 (9.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 95 bytes 9507 (9.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 18 bytes 2112 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 18 bytes 2112 (2.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

**Q.2 Configure a static IP address for your system using nmcli so that your device always gets the same IP (for example, 192.168.1.50/24 with gateway 192.168.1.1 and DNS 8.8.8.8).<br><br><em><strong>Hint:</strong> Use 'nmcli con mod' to set ipv4.addresses, ipv4.gateway, and ipv4.dns, then bring the connection down and up to apply changes.</em>**

**Ans-** Step 1-List Connection: nmcli connection show

Step 2-Restart connection: sudo nmcli connection down ens33

Sudo nmcli connection up ens33

Step 3-Verify: ip add

```
[root@localhost ~]# nmcli connection show
NAME      UUID                                  TYPE      DEVICE
ens33     7cc69e14-d8e1-31df-bc71-521d6a32ff0d ethernet  ens33
lo        0ec17fe9-aa21-4c8c-b609-9957a5cc94b5 loopback   lo
[root@localhost ~]# sudo nmcli connection modify ens33
Error: <setting>.<property> argument is missing.
[root@localhost ~]# sudo nmcli connection modify ens33 \
> ipv4.addresses 192.168.1.50/24 \
> ipv4.gateway 192.168.1.1 \
> ipv4.dns "8.8.8.8" \
> ipv4.method manual
Error: failed to modify ipv4.method: 'manual' not among [auto, link-local, manual, shared, disabled].
[root@localhost ~]# ipv4.method manual
bash: ipv4.method: command not found...
^[A[root@localhost ~]# ^C
[root@localhost ~]# sudo nmcli connection modify ens33
Error: <setting>.<property> argument is missing.
[root@localhost ~]# sudo nmcli connection modify ens33 \
> ipv4.addresses 192.168.1.50/24 \
> ipv4.gateway 192.168.1.1 \
> ipv4.dns "8.8.8.8" \
> ipv4.method manual
[root@localhost ~]# sudo nmcli connection down ens33
Connection 'ens33' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
[root@localhost ~]# sudo nmcli connection up ens33
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
```

```
[root@localhost ~]# ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c6:37:0a brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.1.50/24 brd 192.168.1.255 scope global noprefixroute ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fec6:370a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

**Q.3 Explain in your own words the difference between DNS and DHCP, and give a real-world example of how each is used in an app like Zomato or Flipkart.**

**Ans-DNS (Domain Name System):**

Converts a domain name into an IP address.

Humans remember names like [www.flipkart.com](http://www.flipkart.com), while computers communicate using IP addresses.

Example

When you open [www.flipkart.com](http://www.flipkart.com)



DNS Server



142.xx.xx.xx

- **DHCP (Dynamic Host Configuration Protocol):**

Automatically assigns IP addresses and other network settings to devices.

Example:

When you connect your laptop to Wi-Fi at home:

IP Address

Gateway

DNS Server

Subnet Mask

are automatically assigned by the router.

- **Real-world Example**

- **Zomato**

DNS helps your phone locate [www.zomato.com](http://www.zomato.com).

DHCP gives your phone an IP address when you connect to your home Wi-Fi, allowing it to reach the internet.

**Q.4 Enable the firewall on your system using ufw, then allow incoming connections only on port 80 (HTTP) and block all others. Show the status output after applying these rules.**

**Ans-** Step 1-Check status: `sudo systemctl status firewalld`

Step 2-Enable firewalld: `sudo systemctl enable firewalld`

`sudo systemctl start firewalld`

Step 3- Allow HTTP: `sudo firewall-cmd --permanent --add-service=http`

Step 4-Reload Firewall: `sudo firewall-cmd --reload`

Step 5-Verify: `sudo firewall-cmd --list-all`

```
root@localhost:~ — sudo systemctl status firewalld
[root@localhost ~]# sudo systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset=
   Active: active (running) since Mon 2026-06-29 09:23:38 EDT; 14min ago
     Docs: man:firewalld(1)
    Main PID: 816 (firewalld)
      Tasks: 2 (limit: 10738)
     Memory: 36.9M
        CPU: 2.202s
    CGroup: /system.slice/firewalld.service
            └─816 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Jun 29 09:23:34 localhost systemd[1]: Starting firewalld - dynamic firewall dae
Jun 29 09:23:38 localhost systemd[1]: Started firewalld - dynamic firewall dae
lines 1-13/13 (END)
```

```
[root@localhost ~]# sudo systemctl enable firewalld
[root@localhost ~]# sudo systemctl start firewalld
[root@localhost ~]# sudo firewall-cmd --permanent --add-service=http
success
[root@localhost ~]# sudo firewall-cmd --reload
success
[root@localhost ~]# sudo firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens33
  sources:
  services: cockpit dhcpv6-client http ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

**Q.5 Use firewalld or iptables to temporarily block all outgoing traffic to a specific website (for example, [www.spotify.com](https://www.spotify.com)) and then verify that you cannot ping the site from your terminal.**  
***Hint:*** You may need to use the site's IP address instead of the domain name.

**Ans-**

Step 1-Find spotify ip: ping [www.spotify.com](https://www.spotify.com)

Step 2- Block the ip: sudo firewall-cmd --reload

Step 3-Test: ping [www.spotify.com](https://www.spotify.com)

Step 4-Remove rule: sudo firewall-cmd --reload

```
root@server2:~# ping www.spotify.com
ping: www.spotify.com: Name or service not known
root@server2:~# sudo firewall-cmd --reload
success
root@server2:~#
```